

Threat Hunting in Network Security

Ishani Sharma (2101AI16)

Mamta Kanwar (2101CS42)

Indian Institute of Technology Patna

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1 Introduction

In today's modern digital age, network security has never been as important. While cyber threats have increased in size and numbers, traditional defense protocols like firewalls, antivirus systems, and intrusion detection systems (IDS) alone are not adequate. These post-event systems typically fail to catch sophisticated, hidden threats such as Advanced Persistent Threats (APTs). Hence, there emerged proactive security processes, the most notable among these being threat hunting.

Threat hunting has become an imperative proactive strategy in the realm of cybersecurity, moving beyond conventional reactive security practices to actively hunt for covert enemies within organizational networks. This in-depth examination delves into the methodologies, frameworks, competencies, and practical applications of threat hunting as a vital element of contemporary network security. The study discloses that threat hunting that works is the amalgamation of human insight and deep analytics, which allows organizations to detect and dismantle sophisticated threats that could otherwise evade detection for years or months. Through embracing formalized hunting techniques while ensuring methodological flexibility, security teams are able to tremendously improve their defenses against both new and unknown threats in increasingly intricate digital landscapes.

2 Literature Review

Threat hunting is a paradigm change in cybersecurity strategy from reactive defense to active threat discovery and neutralization. Fundamentally, threat hunting is the act of actively looking for cyber threats that are hidden unseen within a network, having successfully evaded initial endpoint security controls. (1)

2.1 Threat Hunting Fundamentals

- **Proactive approach:** Threat hunting moves from reactive to proactive defense and finding threats that get past initial defenses.
- **Human-centered:** Involves experienced analysts identifying suspect activities outside of automated tools.
- **Assume breach mindset:** Works on the assumption that attackers are already within the network.
- **Reducing dwell time:** Seeks to discover threats early to limit damage from extended intrusions.
- **Behavioral baselining:** Detects anomalies by measuring activity against known good network behavior.
- **Threat intelligence:** Leverages real-time intelligence for hypothesis-driven investigations against stealthy attackers.

2.2 Evolution of Threat Hunting Practices

Threat hunting has become an organized discipline to address the increasingly sophistication of cyber threats and the limitations of traditional, automated defenses. With more sophisticated persistent threats continually evading traditional tools, organizations now depend on human-powered investigation that is able to recognize new threats. Based in humble beginnings with log reviews, threat hunting has matured into a data-driven methodology that incorporates varied sources such as end-point telemetry, network traffic, and threat intelligence. This is a transition from monitoring security passively to proactive detection and neutralization, allowing analysts to detect subtle compromises prior to attackers completing their goals.

2.3 Threat Hunting Methodologies and Approaches

Effective threat hunting initiatives usually use multiple methods, and most tend to use hybrid methodologies that are a combination of various techniques in response to unique organizational requirements and threat environments. Research indicates that threat hunting techniques usually fall under three main categories: hypothesis-driven investigation, investigation based on known indicators of compromise (IoCs) or indicators of attack (IoAs), and custom hunting methods. (2)

- **Hypothesis-Driven Investigation**

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- Actively looks for threats with fresh threat intelligence (TTPs).
 - Has a scientific methodology: formulates hypotheses, tests with data.
 - Needs trained analysts to interpret.
 - Sqrrl Threat Hunting Reference Model is an example that employs a cyclical process.

- **IOC and IOA-Based Hunting**

- Uses known Indicators of Compromise (IoCs) and Indicators of Attack (IoAs).
- Effective for identifying known threats, using CERTs and ISACs.
- Challenges: Very easily manipulated by attackers and not as effective for insider threats.

- **TTP-Based Hunting**

- Centers around adversary Tactics, Techniques, and Procedures (TTPs).
- Conforms to models such as MITRE ATTACK.
- Finds advanced persistent threats and insider attacks.
- Compels attackers to recalculate plans when TTPs are detected.

2.3.1 Threat Hunting Process

The threat hunting process is crucial for actively discovering and countering cybersecurity threats in an organization. The process adopts a formalized methodology, which allows security teams to systematically scan and investigate for possible threats. There are three main stages in the proactive threat hunting process: an initial trigger stage, an investigation, and finally a resolution or escalation to other teams. These stages ensure that hunting processes are targeted, effective, and in line with organizational security goals.

2.3.2 Trigger and Hypothesis Formation

The process begins with collecting information from the environment to formulate hypotheses about potential threats. The trigger is often driven by specific indicators such as:

- New threat intelligence
- Observed anomalies
- Vulnerabilities
- Zero-day exploits

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- Requests from internal teams

During this stage, threat hunters take on the role of hypothetical attackers, thinking about how adversaries might take advantage of vulnerabilities in the organization's environment. This assists in defining the scope and focus of the hunt, guaranteeing the most important areas are probed.

2.3.3 Investigation and Evidence Collection

After hypotheses are created, threat hunters methodically explore the environment to prove or disprove their assumptions. This stage entails:

- Gathering and processing data from a variety of sources including system logs, network activity, application logs, and endpoint telemetry.
- Using solutions such as SIEM, EDR platforms, and UEBA solutions to process the data. (3)

The aim is to collect evidence that confirms or refutes the hypothesis, enabling hunters to verify whether or not a compromise is present.

2.3.4 Pattern and TTP (Tactics, Techniques, and Procedures) Identification

During this stage, hunters concentrate on detecting patterns of malicious activity that match established attack techniques (TTPs). The steps include:

- Correlating and analyzing network activities to reveal concealed attack methods.
- Reconstructing the sequence of events to comprehend how attacks develop.
- Determining the connections between various compromised entities, such as user accounts or third-party services.

Recognition of these patterns aids hunters in recognizing potential threats beforehand and acting to stop further damage.

2.3.5 Resolution and Automation

The last stage of the threat hunting process is:

- Recording the findings, such as the details of what transpired, who was affected, when the events occurred, and where the impact was realized.
- Incorporating the knowledge gained to refine automated detection mechanisms, converting effective manual hunting methodologies to automated rules and queries.

Incorporating these lessons into automated detection systems, security teams can alleviate repeated manual hunting tasks and target developing threats. This stage enables a continuous loop of feedback, improving the defense of the organization in the long term.

Every stage of the threat hunting process guarantees that hunting activities are targeted, effective, and responsive to the security requirements of the organization. The cyclical approach of threat hunting enables ongoing improvement in detecting and mitigating prospective threats, which ultimately improves the organization's capability to respond to cybersecurity incidents rapidly and effectively.

3 Theory

Successful threat hunting involves a versatile toolkit that allows analysts to gather, process, and analyze vast amounts of security data in an efficient manner. Research indicates that a number of categories of tools have become integral parts of contemporary threat hunting programs, such as security information and event management (SIEM) systems, endpoint detection and response (EDR) solutions, managed detection and response (MDR) services, and advanced security analytics platforms(4)

3.1 Key Tools for Threat Hunting

- **SIEM:** Centralizes data gathering, analysis, and correlation across numerous sources, incorporating sophisticated analytics to identify attacks in their initial phases and eliminate false positives.
- **EDR:** Delivers endpoint visibility, utilizing AI and real-time analytics to monitor suspicious activity such as unordinary process relationships or file manipulation.
- **MDR:** Merges expert insight and advanced technology to provide real-time monitoring and proactive threat hunting, delivering industry-specific insights.
- **Advanced Analytics:** Marries big data and AI to detect patterns and anomalies for more in-depth threat insights, enhancing hunting efficiency.

3.2 Role of Machine Learning and AI

- **Automates Analysis:** Machine learning enhances efficiency by automating pattern detection and anomaly recognition in large data sets.

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- **Hybrid Approaches:** Merging ML with frameworks such as MITRE ATTACK enables the prediction of potential threats by identifying behavioral patterns early.

3.3 Threat Intelligence Integration

- **Actionable Intelligence:** Enhances threat hunting by offering context regarding attacker tactics, vulnerabilities, and known threat indicators.
- **STIX/TAXII:** Standardized protocols allow for effective sharing of intelligence, enhancing SIEM functionality and facilitating proactive hunting.
- **External and Internal Intelligence:** Merges external threat feeds with incident data from inside the organization to provide more contextually relevant threat hunting specific to an environment.

3.4 Skills and Capabilities for Successful Threat Hunting

Effective threat hunting demands a wide range of skills and abilities that integrate technical knowledge with analytical mind and good communication. These fundamental competencies allow threat hunters to efficiently gather and analyze security information, detect potential threats, convey their results to stakeholders, and help enhance security posture.

- **Data Analytics:**
 - Proficiency with data science methodologies and analytics tools.
 - Ability to analyze large volumes of security data and extract meaningful insights.
 - Use of data visualization tools to identify patterns and relationships.
- **Pattern Recognition:**
 - Pattern recognition capability of matching attacker methods, tactics, and procedures.
 - Familiarity with normal network traffic to build a baseline.
 - Differentiation of benign anomalies versus possible threats.
- **Forensic Capabilities:**
 - Detailed analysis of possible compromises to learn the how and why.
 - Thorough knowledge of the systems and business processes within the organization.
 - Threat prioritization based on risk and impact to the organization.

- **Communication Skills:**

- Articulating findings clearly, including technical details and business context.
- Translating complex technical information into language that resonates with various stakeholders.
- Communicating recommended action and its potential effect.

- **Collaboration Capabilities:**

- Working with business units and security teams to enhance security.
- Collaboration with incident response teams, vulnerability management, and other security operations.
- Translating findings into tangible security enhancements.

- **Sixth Sense for Threat Hunting:**

- Building an experiential comprehension of threats based on experience and knowledge.
- Having the capability to recognize threats that may not have automated alerts.
- Ongoing practice and learning within real-world scenarios.(5)

4 Frameworks of Threat Hunting

The use of formal models has become more critical for organizations looking to build successful threat hunting programs. A study shows that although 73% of organizations have implemented a formal threat hunting framework, only 38% consistently adhere to it. This disparity points to the difficulty in applying structured methods while still needing the flexibility to be effective hunting. A number of frameworks have appeared to support threat hunting activities, each presenting different views and methodologies for finding and probing potential threats. These frameworks impose structure and consistency on hunting activities while allowing organizations to gauge and enhance their capabilities over time.

- **The Hunting Maturity Model (HMM):**

- Offers a framework for evaluating and building an organization's threat hunting maturity.
- Describes incremental levels of hunting maturity.
- Assists organizations in determining where they are and how to move their capabilities forward.



Figure 1: Threat-Hunting-Maturity-Model

- Takes into account such factors as data gathering, equipment available, processes followed, and level of automation.
- Points out that hunting maturity is associated with automation in hunting processes.
- Points out that even at advanced maturity levels, human skills are still necessary.

- **The Sqr1 Threat Hunting Loop:**



Figure 2: The Sqr1 Threat Hunting Loop

- A four-step process outlining an effective method for hunting.
- Steps include:
 - * Hypothesis generation: Informed guess regarding possible malicious activity.
 - * Tool-supported inquiry: Using technologies to search across gathered information.

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- * Pattern and TTP (Tactics, Techniques, and Procedures) discovery: Identifying malicious activity and attacker methods.
 - * Automated analytics development: Changing successful methods into automated detection function.
 - Focuses on iterative hunting, with each iteration improving on past findings.
- **The MITRE ATT&CK Framework:**
 - Offers a thorough knowledge database of enemy tactics and techniques from actual observations.
 - Puts specific enemy behaviors to known attack patterns.
 - Establishes a shared language for characterizing and classifying threat activities.
 - Aligns threat hunting activity with ATT&CK to effectively determine potential threats holistically.
 - Allows for more efficient communication and coordination regarding threat hunting activity.
 - Enables effective intelligence sharing between organizations via a shared vocabulary.

5 Analysis

5.1 Challenges in Threat Hunting

5.1.1 Data Overload and False Positives

- **Data Overload:** The increasing volume and complexity of data generated across different systems and networks hinder the effective analysis of security incidents. Security analysts are often overwhelmed by vast amounts of logs, network traffic, and endpoint data that require sifting through to identify potential threats.
- **False Positives:** High rates of false positives, which occur when benign activities are incorrectly flagged as malicious, create significant inefficiencies. This leads to alert fatigue, where analysts may miss actual threats or fail to respond promptly. According to a report from *Gartner*, 50% of alerts in a typical Security Information and Event Management (SIEM) system are false positives.
- **Differentiating Malicious from Benign Behavior:** Many internal systems, like administrative or maintenance tasks, can generate patterns resembling malicious activity (e.g., high data

transfers or abnormal logins). Identifying whether such patterns are benign or malicious remains a major challenge for threat hunters. (4)

5.1.2 Skill Gaps and Resource Constraints

- **Skill Shortages:** There is a shortage of skilled cybersecurity professionals capable of performing advanced threat hunting. As reported by *ISC(2)*, the global shortage of cybersecurity professionals stands at 3.1 million.
- **Resource Constraints:** Budget constraints and the high cost of threat hunting tools hinder many organizations from effectively combating cyber threats. The time and effort required for thorough threat investigations often conflict with other organizational priorities.
- **Lack of Clear Metrics:** The effectiveness of threat hunting programs is difficult to measure. There are no universally accepted standards for evaluating threat hunting outcomes, making it challenging for organizations to assess whether their resources are being used efficiently or the impact of their threat hunting efforts.

5.1.3 Organizational and Process Challenges

- **Integration with Security Operations:** Threat hunting teams often struggle with integrating their activities with broader security operations centers (SOCs). Lack of seamless integration results in delays in detecting, investigating, and mitigating threats.
- **Inconsistent Methodologies:** Different teams may follow varied methodologies for threat hunting, leading to inconsistent results. There is a lack of standardization, which impedes collaboration across teams and reduces overall effectiveness.
- **Adapting to Changing Threats:** Threat landscapes evolve rapidly, making it essential for threat hunters to continuously update their techniques. However, many organizations struggle to adapt their threat hunting strategies fast enough to keep pace with new attack vectors, such as those used in sophisticated cyberattacks.

5.2 Threat Hunting in Real Life

- **SolarWinds Hack (2020-2021):**
 - **Overview:** A significant supply chain attack that compromised the Orion software by SolarWinds, affecting thousands of organizations, including U.S. government agencies and private companies. The attackers inserted a backdoor into software updates.

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- **Detection:** The attack was detected through unusual network activity and anomalous log data, marking a case where threat hunting played a key role in identifying the compromise after the initial breach.

- **Emerging Trends in Threat Hunting:**

- **Adoption of Frameworks:** According to a *Forrester* survey, 73% of organizations have adopted or are in the process of adopting threat hunting frameworks like MITRE ATT&CK or Lockheed Martin's Cyber Kill Chain to standardize their threat detection processes.
- **AI/ML Integration:** Machine learning models are becoming essential tools for analyzing large datasets and identifying patterns of anomalous behavior. AI-powered threat hunting tools are capable of identifying subtle threats that would be difficult for human analysts to detect manually.
- **Collaboration Across Organizations:** Many organizations now collaborate and share threat intelligence through platforms like Information Sharing and Analysis Centers (ISACs). This collective approach enables faster identification and response to emerging threats.

5.3 Future Directions in Threat Hunting

The future of threat hunting will be shaped by several key advancements and trends that aim to address current challenges and improve efficiency.(2)

- **Integration of AI and Machine Learning:** AI and ML will become even more integrated into threat hunting, automating routine tasks and enabling better predictive analytics. Machine learning models will assist in recognizing patterns across massive datasets, reducing false positives, and identifying advanced persistent threats (APTs) much faster than traditional methods.
- **Evolution of Hunting Methodologies:** The future will see a shift towards more agile and adaptive hunting methods that blend established frameworks like MITRE ATT&CK with real-time data from ongoing operations. This will allow threat hunters to better respond to dynamic and rapidly evolving cyber threats.
- **Emerging Threat Landscapes:** As cyber threats evolve, focus will shift to addressing specific attack types that are on the rise. Notably, threats such as supply chain attacks, insider threats, and attacks targeting IoT, OT, and cloud environments will dominate. New bespoke techniques, such as advanced evasion tactics, will challenge traditional defense methods.

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- **Increased Automation and Orchestration:** Automated threat hunting will increase efficiency, allowing threat hunters to spend more time on higher-level analysis. The use of orchestration platforms will automate repetitive tasks like data collection and correlation, allowing humans to focus on interpreting complex patterns and decision-making.
 - **Cloud-Native Threat Hunting:** As organizations increasingly migrate to cloud environments, threat hunting will need to adapt to cloud-native architectures. This involves developing specialized methods to detect threats in serverless environments, containers, and across multi-cloud platforms. (3)

5.4 Table: Comparison of Traditional vs. AI-Enhanced Threat Hunting

Aspect	Traditional Threat Hunting	AI-Enhanced Threat Hunting
Data Handling	Manual data collection and analysis	Automated data aggregation and analysis
Detection Speed	Slow, dependent on analyst expertise	Faster, AI models process large data faster
False Positives	High rate of false positives	Reduced false positives through AI models
Resource Efficiency	Resource-intensive and time-consuming	More efficient, reduces need for large teams
Adaptability	Reactive, based on historical data	Proactive, predictive capabilities
Integration	Siloed within security operations	Integrated with broader security ecosystems

Table 1: Comparison of Traditional vs. AI-Enhanced Threat Hunting

6 Conclusion

Threat hunting is essential for contemporary cybersecurity, allowing proactive discovery and containment of advanced threats. It fills gaps in conventional security by presuming breach and proactively hunting for indicators of compromise. It depends on high-quality data, suitable tools, trained people, and support from the organization. In spite of limitations such as data quantity, false positives, and resource limitations, benefits such as earlier detection of threats and lower dwell time make it worth it. (1)

The future of threat hunting will be influenced by AI, machine learning, and bespoke methodologies, and stronger collaboration and intelligence sharing will boost defenses against epidemic threats.

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